Curriculum Vitae

Sina Dadras

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EDUCATION

M.Sc. in Environmental Engineering (Air Pollution) — (23/09/2023 - 23/11/2025)

University of Tehran, Department of Environmental Engineering, Faculty of Environment — Tehran, Iran

GPA: 18.44/20 (Ranked 1st in the program)

Thesis: "Hourly Ozone Prediction Using Ensemble Forecasting to Accurately Capture Peak Concentration: A Case Study of Tehran, Iran"
Advisors: Prof. Khosro Ashrafi (khashrafi@ut.ac.ir), Asst Prof. Majid Shafiepour Motlagh (shafiepour@ut.ac.ir)

Selected Courses:
- Environmental Engineering Fundamentals: **19.25/20**
- Air Pollution Modeling: **19/20**
- New Methods of Measuring Air Pollutants: **19.75/20**
- Geographic Information and Remote Sensing in Air Pollution: **19/20**
- Application of Nano and Biotechnology in Air Pollution Control: **19/20**
- Control of Gaseous Emission Pollutants from Stationary Sources: **19.75/20**

B.Sc. in Civil Engineering — (22/09/2016 - 26/07/2021)

Iran University of Science and Technology (IUST), School of Civil Engineering — Tehran, Iran

GPA: 13.64/20

RESEARCH INTERESTS

Air Pollution Modeling, Physics-Informed Machine Learning (PIML), AI Interpretability & Mechanistic Understanding, Remote Sensing, and Advanced Deep Learning for Environmental Systems.

RESEARCH EXPERIENCE

Air Pollution Forecasting Using Deep Learning and Ensemble Methods (2025-2026)

University of Tehran — Advisors: Prof. Khosro Ashrafi & Asst. Prof. Majid Shafiepour Motlagh

- **Hourly Ozone Forecasting:** Developed the Residual-Aware Meta Ensemble (RAME), a hybrid deep learning framework combining bidirectional recurrent units and dilated convolutions for 1–24-hour ozone concentration forecasting in Tehran. Achieved city-wide mean 24-hour $R^2 = 0.87$, RMSE = 8.12 ppb, and MAE = 5.30 ppb, accurately identifying 96% of critical ozone days.
- **Daily-Averaged $PM_{2.5}$ Prediction:** Designed next-day and day-after-next $PM_{2.5}$ daily average forecasting models based on ensemble learning and meta-learning strategies. The MLP-Meta model achieved RMSE = 6.75 µg/m³ and IOA = 0.922 on Day 1, and RMSE = 7.97 µg/m³ and IOA = 0.878 on Day 2, with the Day 2 early-warning evaluation detecting 75.0% of critical episodes at a false alarm ratio of 16.28%.

PUBLICATIONS

1. Dadras, S., Hajighasemi, E., Ashrafi, K., & Shafiepour Motlagh, M. (2026).
   Hourly Ozone Prediction Using Ensemble Forecasting to Accurately Capture Peak Concentration: A Case Study of Tehran, Iran.
   Environmental Science and Pollution Research.
   https://doi.org/10.1007/s11356-026-38197-7
   Code: github.com/sinadadras021/O3-RAME-forecasting

2. Hajighasemi, E., Dadras, S., Ashrafi, K., & Shafiepour Motlagh, M. (2026).
   Systematic Comparison of Ensemble Learning Strategies for Multi-Day Urban PM2.5 Forecasting.
   Journal of Hazardous Materials Advances (Under Review; joint first author).
   Code: github.com/sinadadras021/PM25-Ensemble-forecasting

HONORS and AWARDS

- **Ranked 1st** in M.Sc. program, Environmental Engineering (Air Pollution), University of Tehran.
- **62nd** of 16,000+ candidates — Civil Engineering Graduate Admission Exam (National).
- **Top 1%** of 150,000+ candidates — National Entrance Exam for Bachelor's Degree.

PROFESSIONAL EXPERIENCE

AI-Driven Industrial Source Attribution & Atmospheric Inversion — University of Tehran & Department of Environment (DOE), Tehran, Iran (Contracted Research Project) — (2025 – 2026)

Developed a physics-informed AI inverse modeling framework to quantify real-time emission variations and source-specific contributions from Tehran's major industrial emitters, including refineries, power plants, and cement facilities. Integrated a regularized Non-Negative Least Squares (NNLS) inversion scheme with boundary-layer meteorology, Briggs dispersion, wind dilution, and thermal-inversion trapping, while leveraging cross-pollutant chemical signatures to non-intrusively identify heavy fuel oil (mazut) switching. The complete pipeline was deployed as a standalone, fully offline desktop GIS application for real-time emission-source attribution and scenario simulation.

Teaching Assistant, Air Pollution Modeling Course — University of Tehran, Tehran, Iran (Academic Appointment) — 2024
Assisted in teaching graduate students the fundamentals of air pollution modeling and Python-based implementation. Developed and delivered tutorials on data-driven modeling techniques, data analysis, and visualization, and provided one-on-one support to strengthen students' understanding of atmospheric dispersion concepts and computational tools.

LANGUAGE and TECHNICAL SKILLS

Languages

- English (C1) — IELTS Overall Band 7.0 (CEFR Level C1) (September 2025)
  Listening: 7, Reading: 7, Writing: 6, Speaking: 7
- Persian: Native

Technical

- Programming & Data: Python (NumPy, Pandas, scikit-learn), TensorFlow, PyTorch; data analysis and visualization; GIS (ArcGIS).
- Scientific Software Practice: Reproducible research code (Git, packaging, configuration-driven pipelines); automated testing against analytical solutions and conservation laws; technical documentation for external readers.
- Air Quality & Modeling: Gaussian plume dispersion modeling; urban-scale ozone and $PM_{2.5}$ forecasting; time-series modeling and validation.
- Quantitative Methods: Statistics for environmental data; calculus and applied mathematical modeling.

REFERENCES

Prof. **Khosro Ashrafi** — M.Sc. Thesis Primary Advisor and Research Supervisor, University of Tehran, Faculty of Environment (khashrafi@ut.ac.ir)

Asst. Prof. **Majid Shafiepour Motlagh** — M.Sc. Thesis Co-Advisor and Research Supervisor, University of Tehran, Faculty of Environment (shafiepour@ut.ac.ir)

Asst. Prof. **Mohsen Mir Mohammadi** — Course Instructor, University of Tehran, Faculty of Environment (mirmohammadi.m@ut.ac.ir)